

REPORT

1. Data Preparation and Preprocessing

The project starts with the `winequality-red.csv` dataset, which contains various physicochemical properties of red wine. The raw data is in a semi-colon separated format, which is properly handled during loading. The goal is to perform a binary classification to predict if a wine is of "good" quality or not. A new binary target variable, `label`, is created from the `quality` column: wines with a `quality` score of 7 or higher are labeled as **1 (Good)**, and those with a score below 7 are labeled as **0 (Not Good)**.

The features (x) are the physicochemical properties (e.g., fixed acidity, volatile acidity, etc.), while the target variable (y) is the newly created binary `label`. The dataset is then split into training (80%) and testing (20%) sets to ensure the model's performance is evaluated on unseen data. Stratification is used during the split to maintain the same proportion of "good" and "not good" wines in both sets, which is crucial for this imbalanced dataset. Finally, the features are scaled using `StandardScaler` to standardize them (mean=0, variance=1), which is a common practice for many machine learning models, including logistic regression, to improve performance.

2. Model Training and Evaluation

A **Logistic Regression** model is trained on the scaled training data. Logistic regression is a statistical model used for binary classification. It models the probability of a certain class or event existing. After training, the model's performance is evaluated on the scaled test data. The evaluation involves two key steps:

- **Prediction:** The model predicts a class label (0 or 1) for each wine in the test set.
- **Probability Prediction:** The model also outputs a probability score for each wine, indicating how likely it is to be of good quality.

The evaluation metrics, including precision, recall, F1-score, and AUC, are calculated from the predictions and probability scores. These metrics are then saved in a CSV file, `metrics_summary.csv`, for future reference.

3. Results Analysis

The evaluation results are presented through a confusion matrix, an ROC curve, and a metrics summary.

Confusion Matrix

The confusion matrix visualizes the performance of the classification model.

- **True Negatives (TN):** 271 "Not Good" wines were correctly classified.
- **False Positives (FP):** 8 "Not Good" wines were incorrectly classified as "Good".
- **False Negatives (FN):** 16 "Good" wines were incorrectly classified as "Not Good".
- **True Positives (TP):** 25 "Good" wines were correctly classified.

The high number of false negatives (16) compared to false positives (8) suggests the model is more conservative in predicting a wine is "good." This could be a result of the class imbalance, where "good" wines are a minority class.

ROC Curve

The ROC (Receiver Operating Characteristic) curve plots the True Positive Rate (TPR) against the False Positive Rate (FPR) at various threshold settings.

The Area Under the Curve (AUC) is a single scalar value that summarizes the ROC curve. An AUC of 0.880 indicates that the model has a high ability to distinguish between "good" and "not good" wines. An AUC of 1.0 represents a perfect classifier, while 0.5 represents a random classifier.

Performance Metrics

The `metrics_summary.csv` file provides a concise overview of the model's performance:

Code snippet

```
precision_macro, recall_macro, f1_macro, auc
0.8023715415019763, 0.6734111325665351, 0.7128075176855665, 0.880362689950466
```

- **Macro Precision (0.802):** This is the average of precision for each class. It shows that when the model predicts a wine is good, it is right about 80% of the time.
- **Macro Recall (0.673):** This is the average of recall for each class. It indicates the model's ability to identify all relevant instances. The relatively lower recall suggests the model struggles more with identifying all of the "good" wines.
- **Macro F1-score (0.713):** This is the harmonic mean of precision and recall. It provides a balanced measure of the model's performance.
- **AUC (0.880):** As discussed above, this confirms the model's strong discriminatory power.

Results:

